

MOSUL, Iraq

WMO: 406080

Lat: 36.306N

Lon: 43.147E

Elev: 216

StdP: 98.76

Time Zone: 3.00 (E03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.9	0.8	-9.8	1.7	6.6	-7.0	2.1	12.4	8.1	9.2	6.7	9.0	0.8	340	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	45.2	20.8	44.1	20.3	43.0	20.2	24.4	38.9	22.8	38.5	21.7	38.2	2.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	15.0	33.8	17.2	12.6	29.2	16.0	11.6	27.3	75.0	39.6	68.5	38.4	64.3	37.9	33.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.1	5.3	-2.7	47.6	1.7	1.3	-4.0	48.5	-5.0	49.3	-5.9	50.0	-7.2	51.0		
			-3.7	25.2	2.2	3.8	-5.2	28.0	-6.5	30.2	-7.7	32.3	-9.3	35.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.7	7.9	10.3	14.3	19.6	26.4	32.6	35.4	35.1	29.7	23.3	15.0	9.9
	DBStd	10.18	2.70	2.98	3.27	3.40	3.48	2.44	2.15	2.02	2.87	3.41	3.24	3.02
	HDD10.0	150	74	29	4	0	0	0	0	0	0	0	2	40
	HDD18.3	1074	323	224	132	23	1	0	0	0	3	107	262	
	CDD10.0	4419	10	39	138	289	507	678	787	778	592	412	152	36
	CDD18.3	2301	0	0	7	62	249	428	529	520	342	157	7	0
	CDH23.3	36592	0	1	84	696	3337	7246	9454	9080	4934	1686	72	3
	CDH26.7	24772	0	0	18	258	1885	5029	7009	6652	3118	797	6	0
Wind	WSAvg	1.9	1.4	1.5	1.7	2.0	2.4	2.7	2.5	2.4	1.9	1.5	1.2	1.2
Precipitation	PrecAvg	476	59	60	65	69	40	12	12	8	11	32	50	59
	PrecMax	634	94	114	133	122	120	37	25	18	33	82	120	119
	PrecMin	285	24	14	27	23	9	3	2	2	1	6	5	13
	PrecStd	84	21	23	27	29	23	9	6	4	7	20	30	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	21.2	28.9	36.8	40.2	44.8	47.2	47.0	43.9	37.1	27.3	23.2
		MCWB	10.7	13.0	14.7	17.1	18.7	19.9	20.8	21.4	20.1	18.0	14.2	12.2
	2%	DB	15.8	19.2	25.2	32.1	38.2	43.1	45.8	45.2	41.2	34.9	25.2	19.0
		MCWB	9.8	11.6	13.8	17.1	18.7	20.0	20.9	20.8	19.3	17.1	13.6	11.1
	5%	DB	14.2	17.8	23.2	29.8	36.8	42.0	44.3	44.1	39.5	33.0	23.8	16.8
		MCWB	9.8	11.1	13.4	15.9	18.3	19.6	20.5	20.4	19.0	17.0	13.6	10.3
	10%	DB	13.0	16.0	21.2	27.2	35.0	40.8	43.2	42.9	38.0	31.2	21.8	15.1
		MCWB	9.4	10.3	12.5	15.4	17.8	19.4	20.2	20.3	18.7	16.3	13.1	10.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	14.2	17.2	19.6	22.8	28.8	26.5	24.2	22.3	20.7	18.4	13.5
		MCDB	15.2	19.2	23.1	28.1	35.6	39.6	40.4	41.8	37.3	32.9	23.1	18.1
	2%	WB	12.0	12.9	15.7	18.4	21.1	25.3	24.0	22.9	21.1	18.9	16.7	12.5
		MCDB	14.6	17.2	21.6	27.3	33.0	37.2	40.5	39.9	36.8	29.8	21.4	16.4
	5%	WB	10.8	11.9	14.5	17.5	20.0	22.8	22.5	22.0	20.3	17.8	15.5	11.7
		MCDB	13.1	16.5	20.8	26.1	31.5	35.6	39.6	39.8	36.0	28.5	19.9	15.6
	10%	WB	9.8	11.0	13.3	16.6	19.0	20.7	21.3	21.1	19.6	17.1	14.1	10.9
		MCDB	12.0	14.8	19.6	24.8	30.8	36.2	39.7	39.9	35.2	28.2	19.5	14.1
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.0	11.4	12.8	14.0	15.2	15.2	15.8	15.6	14.1	12.3	10.7
		MCDBR	11.5	12.3	14.5	16.4	15.9	16.4	16.5	16.5	17.0	16.2	14.6	13.1
		MCWBR	7.0	6.3	6.1	6.0	5.3	5.6	5.7	5.6	5.9	6.5	6.7	7.0
	5% WB	MCDBR	8.6	10.3	12.1	13.6	14.5	15.4	15.0	16.4	15.8	13.6	10.7	10.2
		MCWBR	5.9	5.7	5.9	6.2	5.8	5.8	7.9	6.5	6.6	6.4	6.2	5.9
Clear-Sky Solar Irradiance	taub	0.404	0.431	0.464	0.543	0.534	0.477	0.495	0.467	0.433	0.468	0.402	0.397	
	taud	2.065	2.003	1.941	1.780	1.821	1.948	1.916	1.983	2.063	1.963	2.113	2.096	
	Ebn at Noon	750	782	796	755	769	813	795	809	816	732	748	728	
	Edh at Noon	126	151	175	217	212	186	191	175	153	155	119	115	
All-Sky Solar Radiation	RadAvg	2.44	3.31	4.54	5.59	6.84	8.02	7.69	6.95	5.79	4.08	2.99	2.23	
	RadStd	0.21	0.34	0.34	0.36	0.39	0.37	0.22	0.24	0.27	0.30	0.28	0.28	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page